

LAMPNODE · DESIGN DECISION RECORD

LampNode recommendations accepted

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DOCUMENT CONTROL

Document ID	LPN-DDR-002	Version	0.2
Status	Draft	Date	2026-10-02
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/lampnode		
Source	docs/decisions/0002-recommendations-accepted.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-10-02	Amish Chadha	O1 (first partner) decided and the pinout to propose to FieldNode recorded, 2026-10-02
0.1	2026-09-25	Amish Chadha	Record Amish's acceptance of all recommendations in LPN-DDR-001 and docs/REVIEW.md, what changed in the repo, and the items still open

0002: Recommendations accepted

- **Date:** 2026-09-25
- **Status:** accepted. Every item with a recommendation is decided by Amish; one item without a recommendation stays open.

■ Context

LPN-DDR-001 v0.1 and the TRL 3 session of `docs/REVIEW.md` held ten items adopted for TRL 3 work, open for Amish's review (D1 to D10), and eight open items (O1 to O8), seven of which carried a recommendation or an engineering proposal. On 2026-09-25 Amish wrote: "i accept all your recommendations, go with them across all repos." Every item with a recommendation is therefore **decided by Amish, 2026-09-25: go with recommendation**. Where a recommendation offered several options, the recommended option is the decision. Items with no recommendation stay "Proposed, awaiting Amish". TRL 4 remains on hold by Amish's instruction, so decisions that need building, testing or purchasing are recorded but not carried out.

■ Decision

Table 1. Items decided by Amish on 2026-09-25 and what changed in the repo.

#	Decision	What changed in the repo
D1	ANSI C136.41 7-contact socket first; Zhaga Book 18 as a later variant	Status wording only; the model already uses the C136.41 base
D2	24 GHz Doppler radar; schedule-only kept as a firmware mode	Status wording only
D3	Cable from the controller to a head clamped under the arm	Status wording only; geometry unchanged
D4	LoRaWAN on the STM32WL-class module shared with FieldNode and TwinKit	Status wording only
D5	0 to 10 V only; DALI-2 D4i only if a partner's stock needs it	Status wording only; R4 already redefined in LPN-REQ-001 v0.3
D6	Fail-on, normally closed relay, coil energized in daytime (0.203 W average)	Status wording only
D7	Sibling sensors stay on FieldNode solar; LampNode ports optional for them; hosted sensors on cabinet-switched feeders bring storage	Status wording; wording in AirStreet and NoiseMap listed as a cross-repo action
D8	Ships at photocell-equivalent full output; the road owner sets any dimming	Status wording only
D9	120 to 277 V now; 347 V and 480 V deferred until a partner needs them	Status wording only; R2 already redefined
D10	A TALQ bridge is CityTwin scope	Status wording; listed as a cross-repo action for CityTwin
O2	R7 shortfall: option (c), keep the 35 % target and the reference profile and accept R7 at risk until real traffic counts exist	R7 target unchanged (35 %); status stays at risk (34.1 to 36.4 %); LPN-REQ-001 v0.4 and LPN-CAL-001 v0.2 note the decision
O3	R13 at high temperature: option (a), firmware limits the two ports to 2.5 W in total when the interior is above 50 °C (no 10 W supply)	R13 target restated in LPN-REQ-001 v0.4; LPN-CAL-001 v0.2 section 11 and <code>sizing.py</code> add the limited case: peak 3.90 W against 3.55 W becomes 3.40 W against 3.57 W at 64.3 °C; R13 moves from at risk to met on paper; firmware rule added to LPN-PRC-001 v0.4. No BOM, geometry or drawing change
O4	TCXO real-time clock (+\$3)	Confirmed; already in <code>bom/bom.csv</code> line 7; wording updated in <code>bom/bom-notes.md</code>
O5	Normally closed relay rated 80 A inrush or better, closed at the zero crossing (+\$2)	Confirmed; already in <code>bom/bom.csv</code> line 5. Checking the rating against real driver inrush is TRL 4 work, decided but on hold
O6	385 V class varistor and TVS diodes, no gas discharge tube (+\$1)	Confirmed; already in <code>bom/bom.csv</code> line 3. Surge testing is TRL 4, on hold
O7		Confirmed in LPN-PRC-001; firmware beyond a sketch and bench checks are TRL 4, on hold

#	Decision	What changed in the repo
	Charge-pump coil drive, day decision needing both clock and light sensor, I/Q radar	
O8	5-pole M12 expansion ports to match FieldNode	Confirmed; already in <code>bom/bom.csv</code> lines 10 and 14. The pinout is a cross-repo action with FieldNode (its O2)

No recommendation changed the budget, the pitch or the problem, so `budget_usd` stays at 150 and the pitch and problem in `project.yaml` and `README.md` are unchanged. The BOM total stays at \$130.00. `trl` and `trl_target` stay at 3.

■ Items still open

Table 2. Proposed, then decided by Amish on 2026-10-02.

#	Item	Status
O1	First partner (city, utility or campus) and region for co-design	No recommendation was made. Decided by Amish, 2026-10-02 (LPN-DEC-001): a US partner on the 915 MHz band; first candidate to approach: a university campus or municipal utility in the Dallas-Fort Worth area that runs LED streetlights with ANSI C136.41 sockets

■ Cross-repo actions

These are recorded here and in `docs/REVIEW.md` ; no other repo was edited.

- **FieldNode:** agree the 5-pole M12 port pinout (FieldNode O2) and the supply voltage on the ports (FieldNode offers 3.3, 5 or 12 V at build; LampNode offers 12 V only). Decided 2026-10-02 (LPN-DEC-001): LampNode proposes to FieldNode pin 1 supply (12 V on LampNode), pin 3 ground, pins 2 and 4 a two-wire RS-485 pair, pin 5 a wake line; hosted sensors accept 5 to 12 V and regulate down themselves.
- **AirStreet and NoiseMap:** mention LampNode 12 V ports as an optional power source, keeping FieldNode solar as the default (D7).
- **CityTwin:** add a TALQ bridge to CityTwin scope (D10).

■ Consequences

- Requirement status: 0 not met, 4 at risk (R3, R6, R7, R11), 11 met on paper or by design review, 1 not verifiable at TRL 3 (R1). Before: 5 at risk and 10 met.
- Hosted sensors must accept 2.5 W in total, not 3 W, on hot days.
- Controlled documents changed: LPN-DDR-001 v0.2, LPN-REQ-001 v0.4, LPN-PRC-001 v0.4, LPN-PRB-001 v0.4, LPN-CAL-001 v0.2.
- TRL 4 work (parts selection against datasheets, bench build, tests, firmware) stays on hold.